James Pham

jamespham2000@gmail.com | 780-720-2916 | jamespham.com | James Pham | jamespham0317

Education

University of British Columbia *Vancouver, BC*
Bachelor of Science (Computer Science) — 3.95 / 4.33 GPA *Expected Graduation: 2027*

Awards: Trek Excellence Scholarship (Top 5% GPA in undergraduate year), Dean's Scholar

Technical Skills

Programming Languages: JavaScript/TypeScript, Python, C/C++, SQL, Java, HTML/CSS, R, Bash
Full-Stack Development: React, Node.js, Express.js, Next.js, REST API
Technologies: Git/GitHub, AWS, PostgreSQL, TensorFlow, NumPy, Pandas, Scikit-Learn, Matplotlib, JUnit

Experience

Software Developer *Vancouver, BC*
UBC Data Science Club *September 2025 – Present*

- Collaborating in an **AGILE** development environment with weekly team meetings to design and develop the club's primary **website portal** using **React** and modern front-end frameworks
- Partnering closely with the **design team** to translate mockups and user feedback into responsive, accessible, and visually consistent web interfaces
- Assisting in the deployment and maintenance of scalable cloud-based applications on **AWS** and integrating **REST APIs** to enable efficient communication between front-end and back-end services

Projects

Thyme Saver Full-Stack Web Application — *React, Node.js, AWS* *March 2025*

- Designed and developed a **full-stack** AI assistant using **React** and **Node.js**, integrating the **Google Gemini Vision API** to deliver an intuitive and dynamic user experience
- Designed and implemented a **RESTful API** using **Node.js** and **Express** to handle user authentication and secure image uploads via **Multer**, enabling seamless integration between the frontend and backend
- Enhanced security by implementing user login-credential **encryption** via **Bcrypt** and securely stored authentication data in a **PostgreSQL** database hosted on **AWS RDS**
- Deployed the frontend to **AWS S3** and the backend using **AWS Lambda** behind **API Gateway**, with **CI/CD** pipelines via **GitHub Actions** to automate testing, packaging, and deployment - reducing deployment time by 80% and ensuring automated delivery within 2 minutes of each push

Tumor Classification via CNN — *Python, TensorFlow, Matplotlib* *May 2025*

- Engineered a **convolutional neural network (CNN)** pipeline on over 7,900+ images to classify histopathology images into benign and malignant classes, utilizing **Conv2D**, **MaxPooling2D**, and dense layers with **ReLU** and **sigmoid activations**
- Integrated **ImageDataGenerator** for real-time image augmentation and implemented **EarlyStopping** with validation monitoring to prevent **overfitting** and restore optimal model weights, achieving 87% training accuracy and 82% validation accuracy

Maze Generation and Solving Visualization Tool — *C++, Raylib* *April 2025*

- Developed an interactive maze generation and solving **visualization tool** using **C++** and the **Raylib graphics library**, supporting real-time animation and performance metrics at 60 FPS
- Implemented various generation and solving **algorithms** including A* search and Kruskal's algorithm, allowing for efficient route calculation through procedurally generated mazes of variable size and complexity

Leadership and Awards

Mentor *Edmonton, AB*
You Can Ride 2 *March 2023 - April 2023*

- Volunteered as a **mentor** in a six-week program, assisting children with mental challenges and disabilities in building the skills and confidence to ride a bike independently

Telus Friendly Future Social Impact Award *April 2023*

- Awarded $5,000 for displaying social impact initiative